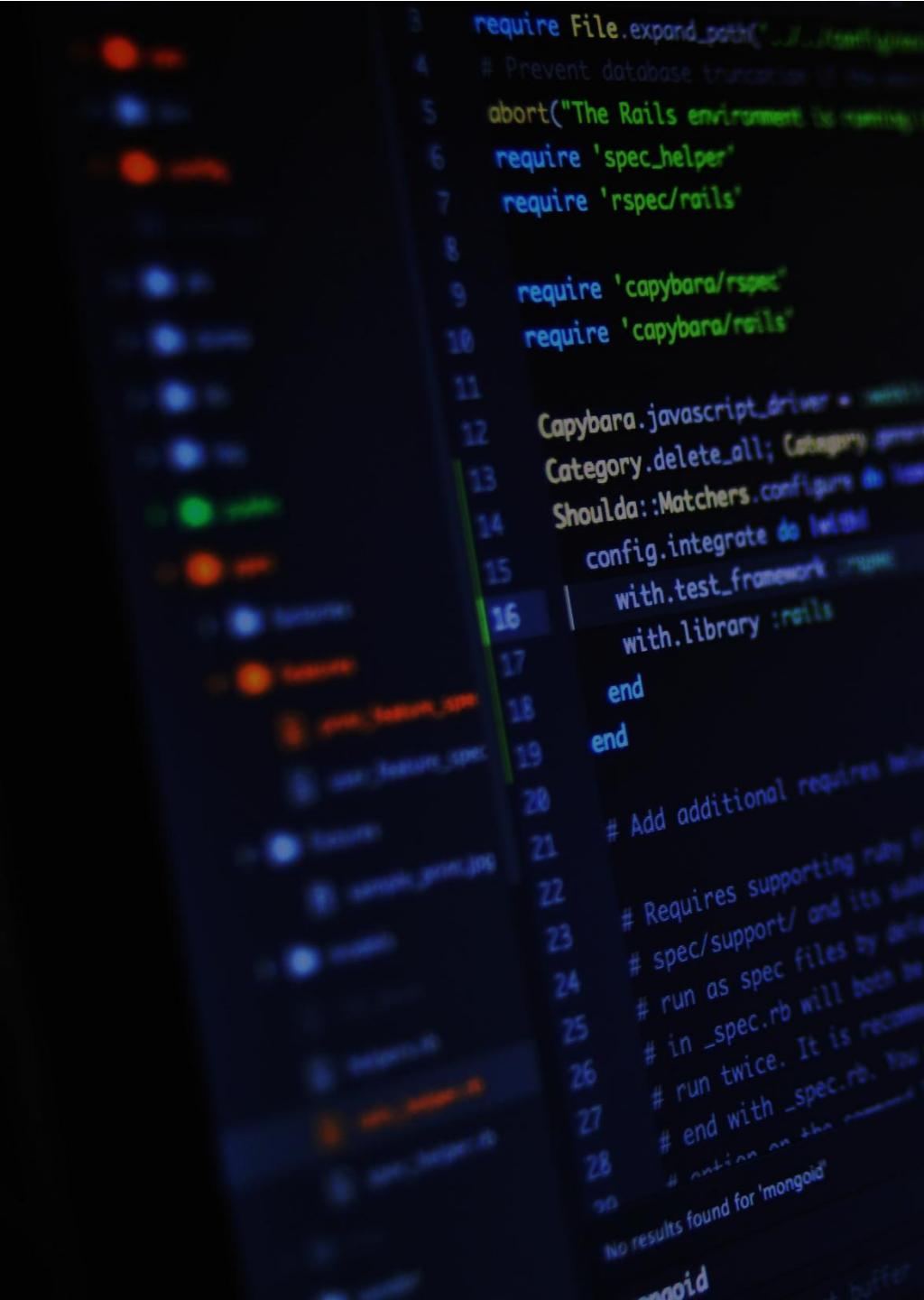




LECTURE 9

configuration management Tools Git / GitHub

Web development 2

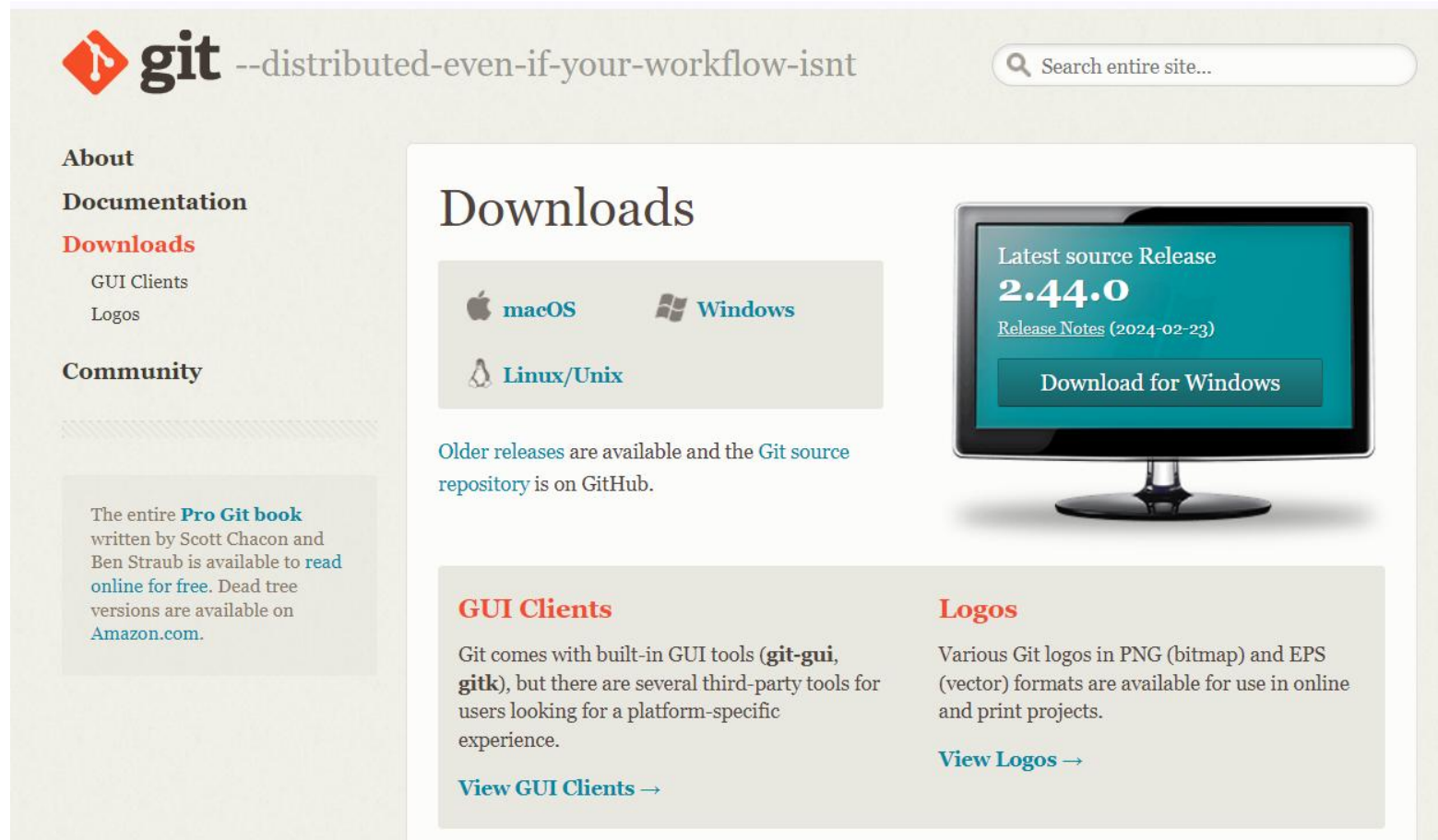
What is Git

- Tool for version control system
help for following working directory changes and simplify return to any version of the project versions.
- Every Commit consider as a version of project versions
- Help for working in a teamwork with your Team through (Branches)

How To install Git

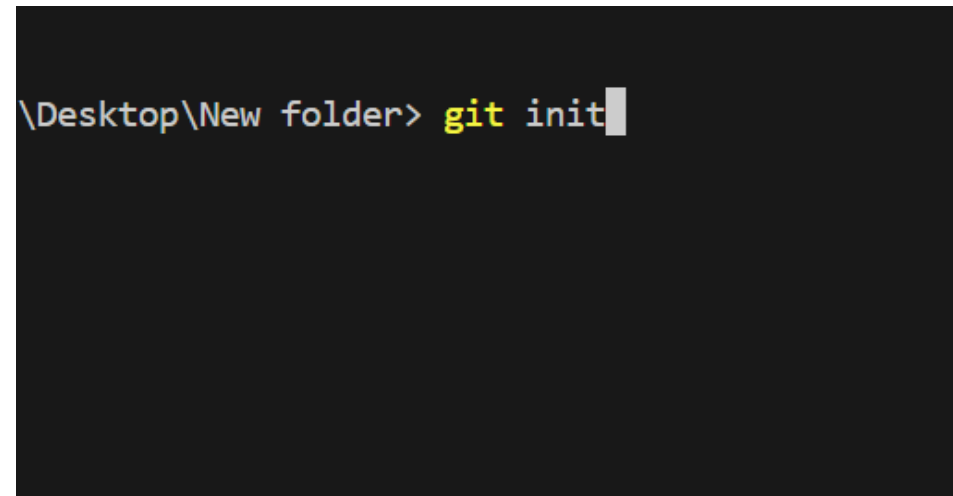
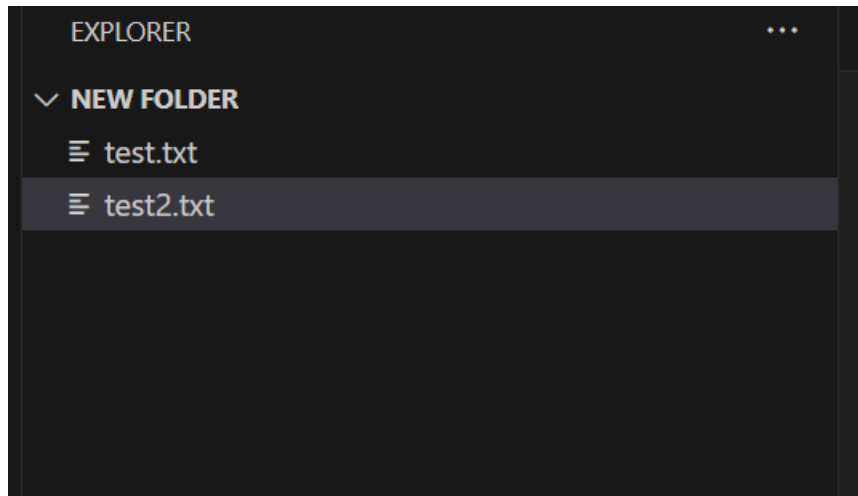
Go to git
Official website
And select your
operating system
then download

After install it
Check installation
Trough
Write
Git –version
Into terminal



Create Git repo

- Create new folder with some files which called working Directory
- Into terminal write **Git init**



This command will create a new hidden file named .git, into this file git will store all project versions and configurations

Check repo status

- Using command git status you can check your project status

```
PS C:\Users\Sultan Gdn\OneDrive\Desktop\New folder> git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        test.txt
        test2.txt

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\Sultan Gdn\OneDrive\Desktop\New folder> 
```

Check repo status

With this command you will find out

- The version or the last commit
- The all untracked files (the files which git doesn't track any changes into it)
- To add it to be tracked by git we use
 `git add <file-name>`
or
 `git add .`
Or
 `git add *`
To add all available files to be tracked

Track files

After track files , git will track all changes into these files

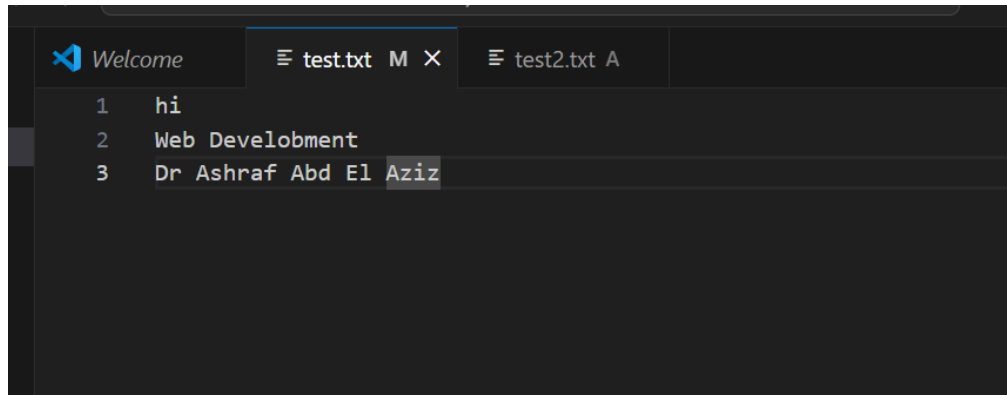
```
nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\Sultan Gdn\OneDrive\Desktop\New folder> git add .
PS C:\Users\Sultan Gdn\OneDrive\Desktop\New folder> █
```

Now we tracked all files

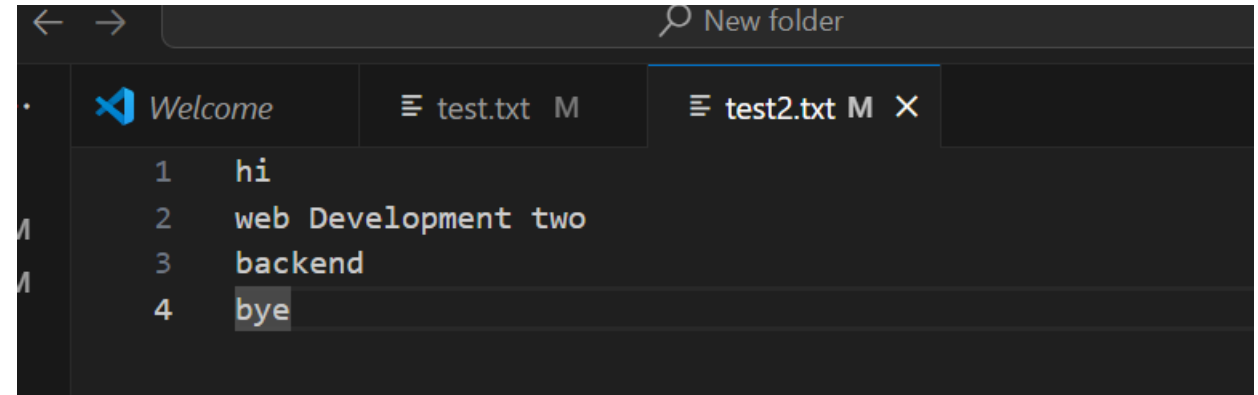
note: to track one file we use git add <file-name>

Check repo status

Now when change test.txt



A screenshot of the Visual Studio Code editor interface. The top bar shows the 'Welcome' tab and two open files: 'test.txt' (modified) and 'test2.txt' (added). The 'test.txt' file is active and contains three lines of text: '1 hi', '2 Web Development', and '3 Dr Ashraf Abd El Aziz'. The third line is highlighted with a mouse cursor.



A screenshot of the Visual Studio Code editor interface. The top bar shows the 'Welcome' tab and two open files: 'test.txt' (modified) and 'test2.txt' (modified). The 'test2.txt' file is active and contains four lines of text: '1 hi', '2 web Development two', '3 backend', and '4 bye'. The fourth line is highlighted with a mouse cursor.

Then using git status to check changes into these files

Check repo status

```
PS C:\Users\Sultan Gdn\OneDrive\Desktop\New folder> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   test.txt
    new file:   test2.txt

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   test.txt
    modified:   test2.txt

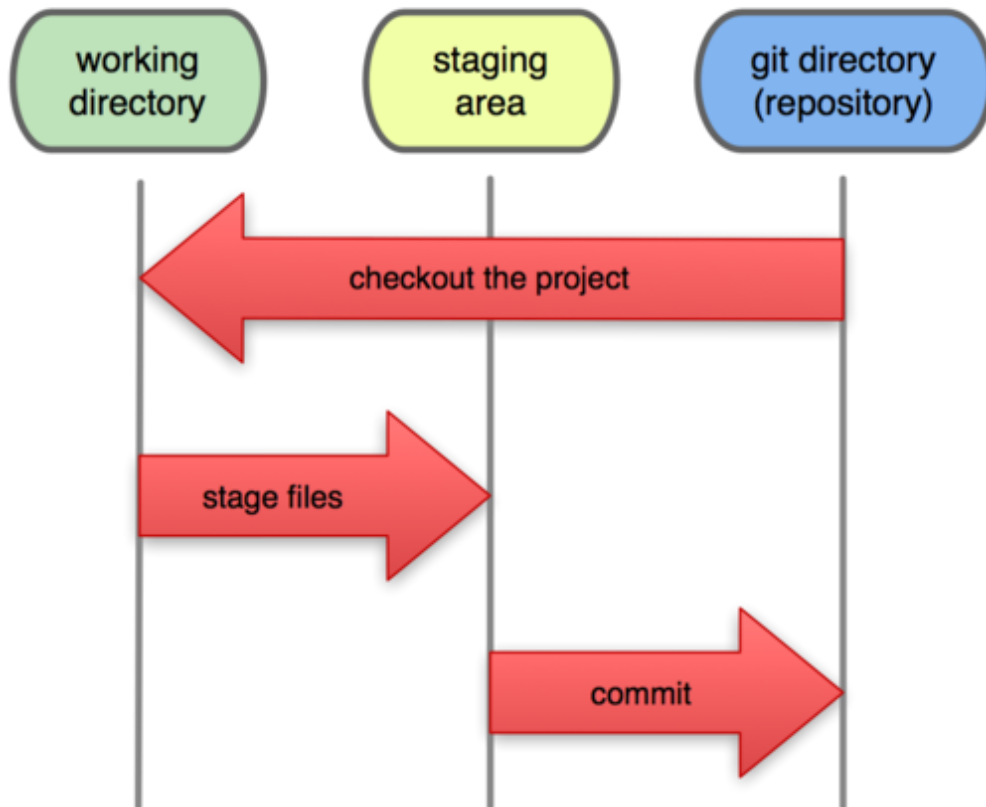
PS C:\Users\Sultan Gdn\OneDrive\Desktop\New folder> █
```

We will notice that git detect that there are new files added test.txt and test2.txt

and detect that there are new changes into these files in another word (**staged**) files

The three stage Architecture

Local Operations



Finally To save the last status
and you must add message to save this status
With command

Git add . (to add the last modifies)

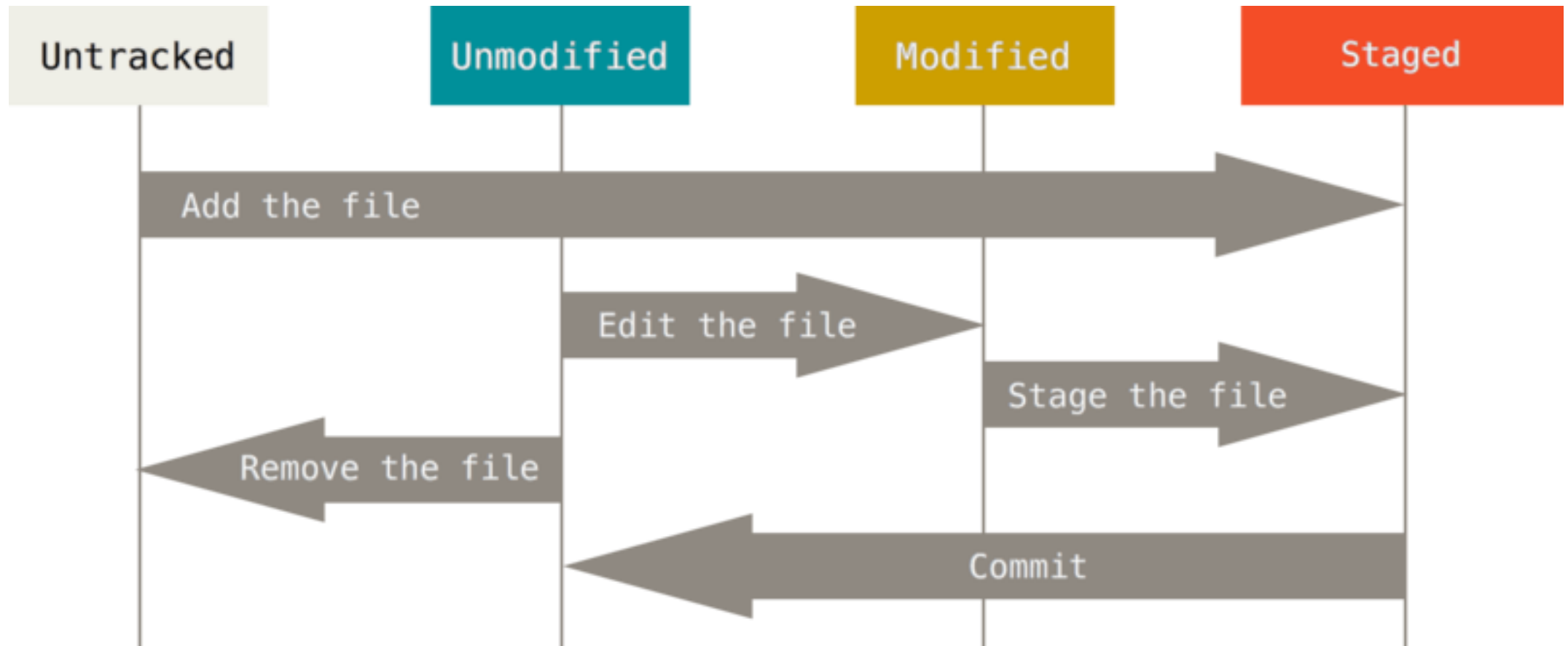
Git commit -m "message"

Then we write git status to notice

```
PS C:\Users\Sultan Gdn\OneDrive\Desktop\New folder> git commit -m "first commit"
[master (root-commit) 15b0d2f] first commit
2 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 test.txt
create mode 100644 test2.txt
PS C:\Users\Sultan Gdn\OneDrive\Desktop\New folder> |
```

Note: we only making new commit after complete each new feature

File status into git repo



File status into git repo

- Git divide all files to tracked and untracked
- We use `git add .` To change file status to tracked file
- Tracked file has some status
unmodified , modified and staged
- So we notice that `git add .` , change file status from untracked to tracked , and from modified to staged
- Then we use `git commit` to change file status from staged to unmodified again.

Another commend lines

- Git status -s
show status in simple view
?? Untracked
M modified
or nothing if stable or unmodified version
- Clear,
is a commend to clear existing commends into terminal

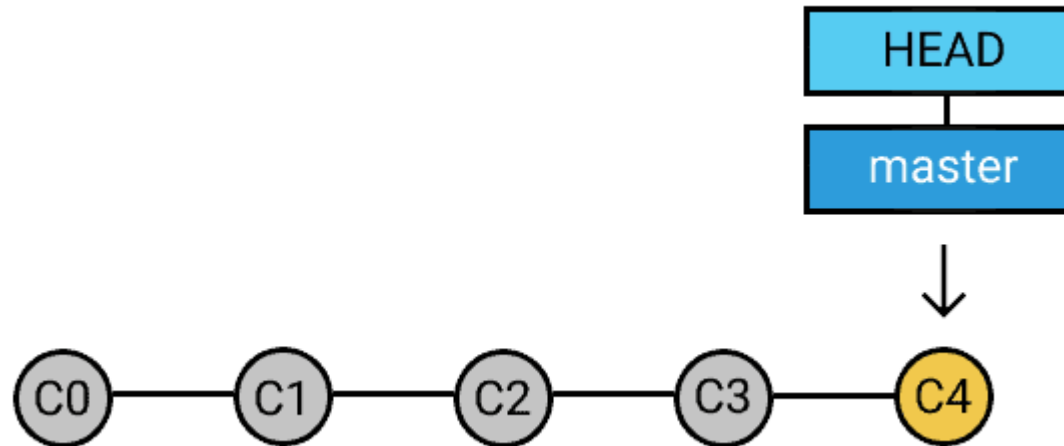
Repo History

- You can find out all all your project history using git log.
- And you can exit git log using Q.
- Each commit has unique ID, Or Hash Number of this commit.
- You can Go back to any version or commit using git checkout <HASH-NUMBER>
And HEAD will change to this commit

What is Git Head

"HEAD" in Git is a pointer to the latest commit in the branch you're currently working on. It's like a bookmark indicating where you are in the project's history.

Note that the last commit named Master Or main



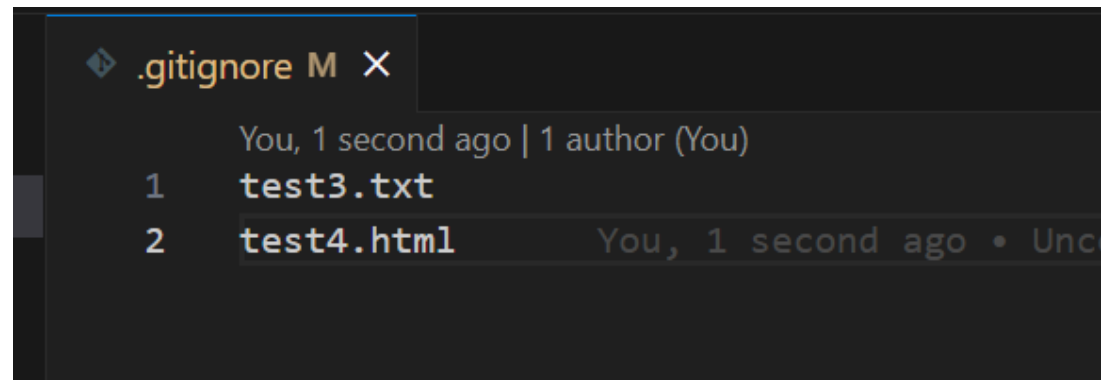
After checkout to commit C3 Head will be on C3 and so.

Git ignore

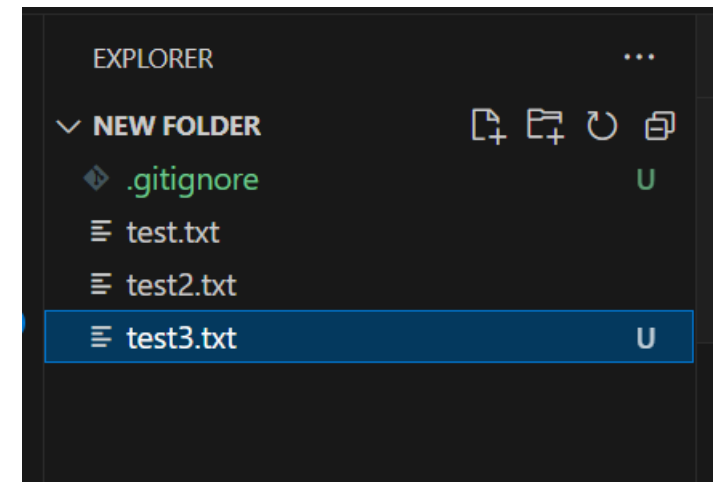
- Why git ignore?
You will need no ignore any changes into some files or directories for example files with security issues or directories and files with large size
- If you already Added file to be tracked you can remove it from git cache to ignore any changes using
`git rm --cached <File-Name>`
- So now this file status will change to be untracked

Git ignore

- To preserve keeping this file untracked and ignore any changes we will create file named .gitignore
- Add all files name you want to ignore it into this files
- You can ignore group of files with specific extension like all txt files by ignore
*.txt

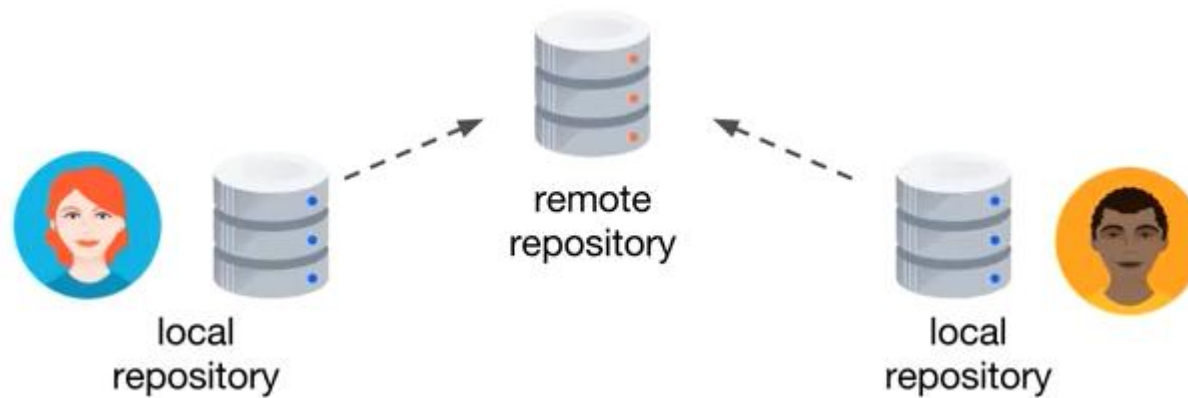


A screenshot of a code editor window titled ".gitignore M X". The editor shows two lines of text: "1 test3.txt" and "2 test4.html". Above the first line, it says "You, 1 second ago | 1 author (You)". To the right of the second line, it says "You, 1 second ago • Unco".



What is the difference between git and github

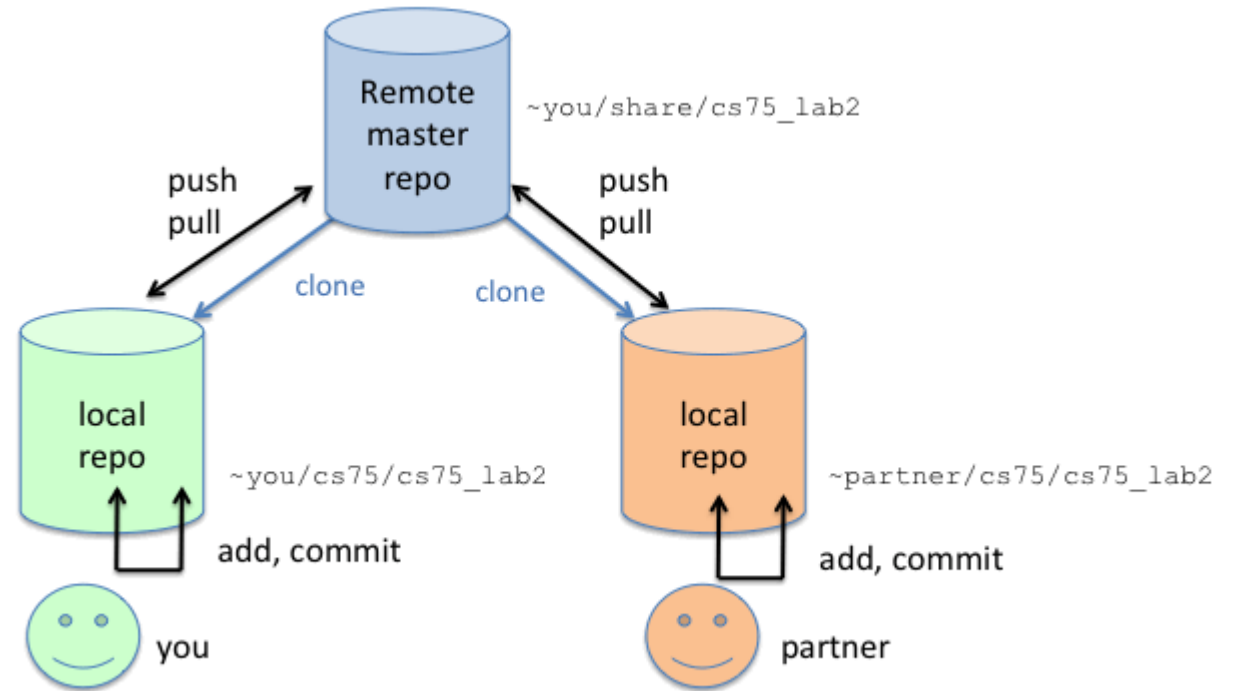
- Any changes or different versions of your project are in your local device Exclusive.
- In case of working with partner or development team we will create central **remote** repository



- After creating local repo , you should create remote repo

What is the difference between git and github

- After any changes or version into your local repo you will push this version, commit to your partner
- Your partner should pull this commit



What is the difference between git and git-up

- The physical location which will host your remote repo called **GitHup**
- Note : You have alternative choices like git-lap or bitbucket



GitLab

ATLASSIAN



Bitbucket



GitHub

Create remote repo and push & pull

- **We** can create remote repo using
git remote add <repo-link>
- And use
git remote -v to check it

resources

- <https://www.w3schools.com/>